.NET Web Development Master Interview Compendium (v2)

A complete 600-question technical study guide spanning Junior, Mid-Level, Senior, and Enterprise Architect roles across C# 1 through C# 15, .NET 10 LTS, .NET 11 STS, .NET Aspire, and Modern Architecture.

LEVEL 1: JUNIOR .NET DEVELOPER

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Answer: Auto-implemented properties allow brief property syntax (`public string Name { get; set; }`) where the C# compiler generates a hidden private backing field automatically.

Q138: What is Single-File C# Apps in .NET 10 in modern .NET development?

Answer: Starting in .NET 10, runnable single-file C# scripts (`main.cs`) execute directly via `dotnet run main.cs` without requiring a `.csproj` or `.sln` file, referencing NuGet packages via inline `#:` directives.

Q139: What is Delegates in C# in modern .NET development?

Answer: A delegate is a type-safe function pointer holding references to methods matching a specific signature (`delegate void ProcessHandler(int id);`).

Q140: What is Action and Func Delegates in modern .NET development?

Answer: System.Action represents delegates that return void (up to 16 parameters). System.Func represents delegates that return a generic type parameter `TResult`.

Q141: What is Events and Publisher-Subscriber in modern .NET development?

Answer: Events encapsulate delegates using the `event` keyword, allowing classes to notify registered subscribers when an action occurs while preventing external code from clearing the invocation list.

Q142: What is Lambda Expressions in modern .NET development?

Answer: Lambdas (`x => x * 2`) are anonymous functions used to construct inline delegates or Expression Trees, commonly passed into LINQ extension methods.

Q143: What is Dependency Injection Service Lifetimes in modern .NET development?

Answer: ASP.NET Core DI provides three lifetimes: Transient (new instance per request), Scoped (created once per HTTP request context), and Singleton (created once for the application lifetime).

Q144: What is Minimal APIs vs Controller APIs in modern .NET development?

Answer: Minimal APIs map HTTP endpoints directly using lambdas on `WebApplication`, bypassing controller instantiation overhead for lightweight, high-performance microservices.

Q145: What is EF Core DbContext & Migrations in modern .NET development?

Answer: DbContext acts as the unit of work and repository session in EF Core. Migrations translate entity model updates into SQL schema scripts using `dotnet ef migrations add`.

Q146: What is RFC 7807 ProblemDetails in modern .NET development?

Answer: ASP.NET Core exception-handling middleware converts unhandled exceptions into RFC 7807-compliant `ProblemDetails` JSON responses containing title, status, instance, and trace identifiers.

Q147: What is Nullable Reference Types (NRTs) in modern .NET development?

Answer: Enabled via `#nullable enable`, NRTs provide compile-time warnings when reference variables might contain null, reducing runtime `NullReferenceException` occurrences.

Q148: What is Null-Coalescing Operators in modern .NET development?

Answer: The `??` operator returns the left-hand operand if non-null; otherwise, the right operand. `??=` assigns the right operand only if the left operand is currently null.

Q149: What is LINQ Deferred Execution in modern .NET development?

Answer: LINQ queries over `IEnumerable` or `IQueryable` execute lazily when enumerated (e.g., in a `foreach` loop or via `.ToList()`), allowing dynamic query composition before execution.

Q150: What is Generics and Constraints in modern .NET development?

Answer: Generics allow classes and methods to operate on placeholder types `T`. Constraints (`where T : class, new()`) enforce structural or interface rules on `T` at compile time.

LEVEL 2: MID-LEVEL .NET DEVELOPER

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Answer: Distributed caching stores serialized state across multiple web server instances using Redis via `IDistributedCache`, ensuring session consistency across load-balanced farms.

Q276: What is Span and ReadOnlySpan in modern .NET development?

Answer: Span represents a contiguous, type-safe region of arbitrary memory (stack, heap, or unmanaged) without allocating new objects or array copies. C# 14 adds implicit span conversions.

Q277: What is C# 14 Extension Everything in modern .NET development?

Answer: C# 14 introduces extension blocks allowing developers to declare instance and static extension properties, extension methods, static extension methods, and operators on existing types.

Q278: What is HybridCache in ASP.NET Core in modern .NET development?

Answer: HybridCache provides a unified, zero-boilerplate API combining fast L1 in-memory caching with L2 distributed caching (e.g., Redis), featuring built-in stampede protection against thundering herds.

Q279: What is IQueryable vs IEnumerable in modern .NET development?

Answer: IEnumerable executes in-memory filtering over object collections in the CLR process. IQueryable parses Expression Trees into native SQL executed on the remote database server.

Q280: What is Passkey Authentication in .NET 10 in modern .NET development?

Answer: ASP.NET Core 10 Identity natively integrates FIDO2/WebAuthn passkeys, enabling passwordless login using biometric hardware (TouchID/Windows Hello) or security keys.

Q281: What is Testcontainers.NET for Integration Tests in modern .NET development?

Answer: Testcontainers.NET programmatically provisions real Docker containers (SQL Server, PostgreSQL, Redis, Kafka) during test execution, eliminating behavioral differences caused by in-memory stubs.

Q282: What is ASP.NET Core Middleware Pipeline in modern .NET development?

Answer: Middleware components form an HTTP pipeline using `Use`, `Run`, and `Map`. Requests pass sequentially through registered middleware delegates before returning responses.

Q283: What is gRPC vs SignalR vs REST in modern .NET development?

Answer: REST is universal for web clients over HTTP. gRPC provides high-performance binary Protocol Buffers over HTTP/2 for inter-service communication. SignalR enables real-time bi-directional WebSockets streaming.

Q284: What is .NET Aspire Core Pillars in modern .NET development?

Answer: .NET Aspire simplifies cloud-native app development through three pillars: 1) Orchestration (AppHost for service lifecycles), 2) Integrations (resilient NuGet components), and 3) Observability (OpenTelemetry dashboard).

Q285: What is Dapper vs EF Core in modern .NET development?

Answer: EF Core is a full ORM supporting change tracking and migration generation. Dapper is a micro-ORM executing raw SQL queries directly into object mappings with maximum performance and minimal memory overhead.

Q286: What is CancellationTokens in Async Methods in modern .NET development?

Answer: CancellationToken propagates cancellation signals through asynchronous call chains, allowing long-running operations or HTTP requests to abort work gracefully when clients disconnect.

Q287: What is Async/Await State Machines in modern .NET development?

Answer: The C# compiler rewrites async methods into struct state machines. Awaiting a non-completed task yields control back to the caller without blocking thread execution.

Q288: What is ConfigureAwait(false) Usage in modern .NET development?

Answer: Calling `ConfigureAwait(false)` prevents the async continuation from capturing the SynchronizationContext, avoiding deadlocks in legacy UI contexts and improving performance in library code.

Q289: What is ValueTask vs Task in modern .NET development?

Answer: Task allocates a reference object on the heap. ValueTask is a discrimination struct used when async methods complete synchronously or return cached values, avoiding heap allocations.

Q290: What is EF Core Query Optimization in modern .NET development?

Answer: EF Core query performance is optimized by using `.AsNoTracking()` for read-only queries, index tuning, compiled queries, and avoiding N+1 query execution through `.Include()`.

Q291: What is JWT Bearer Token Architecture in modern .NET development?

Answer: JWTs contain Header, Payload (claims), and Signature segments. ASP.NET Core validates token signatures, expiration dates, and audience claims via `JwtBearerHandler` middleware.

Q292: What is OAuth 2.0 and OpenID Connect (OIDC) in modern .NET development?

Answer: OAuth 2.0 is an authorization framework issuing access tokens. OpenID Connect is an identity layer built on OAuth 2.0 that issues ID tokens for user authentication.

Q293: What is Action Filters vs Middleware in modern .NET development?

Answer: Middleware operates globally on raw HttpContext. Action Filters execute inside the MVC routing pipeline before and after action execution, accessing action arguments and model state.

Q294: What is FluentValidation Integration in modern .NET development?

Answer: FluentValidation decouples validation rules from request models, using a fluent C# builder interface (`RuleFor(x => x.Email).EmailAddress()`) executed during pipeline execution.

Q295: What is BackgroundService and IHostedService in modern .NET development?

Answer: BackgroundService is a base class implementing `IHostedService` that executes long-running background worker tasks inside ASP.NET Core application hosts.

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LEVEL 4: ENTERPRISE ARCHITECT

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